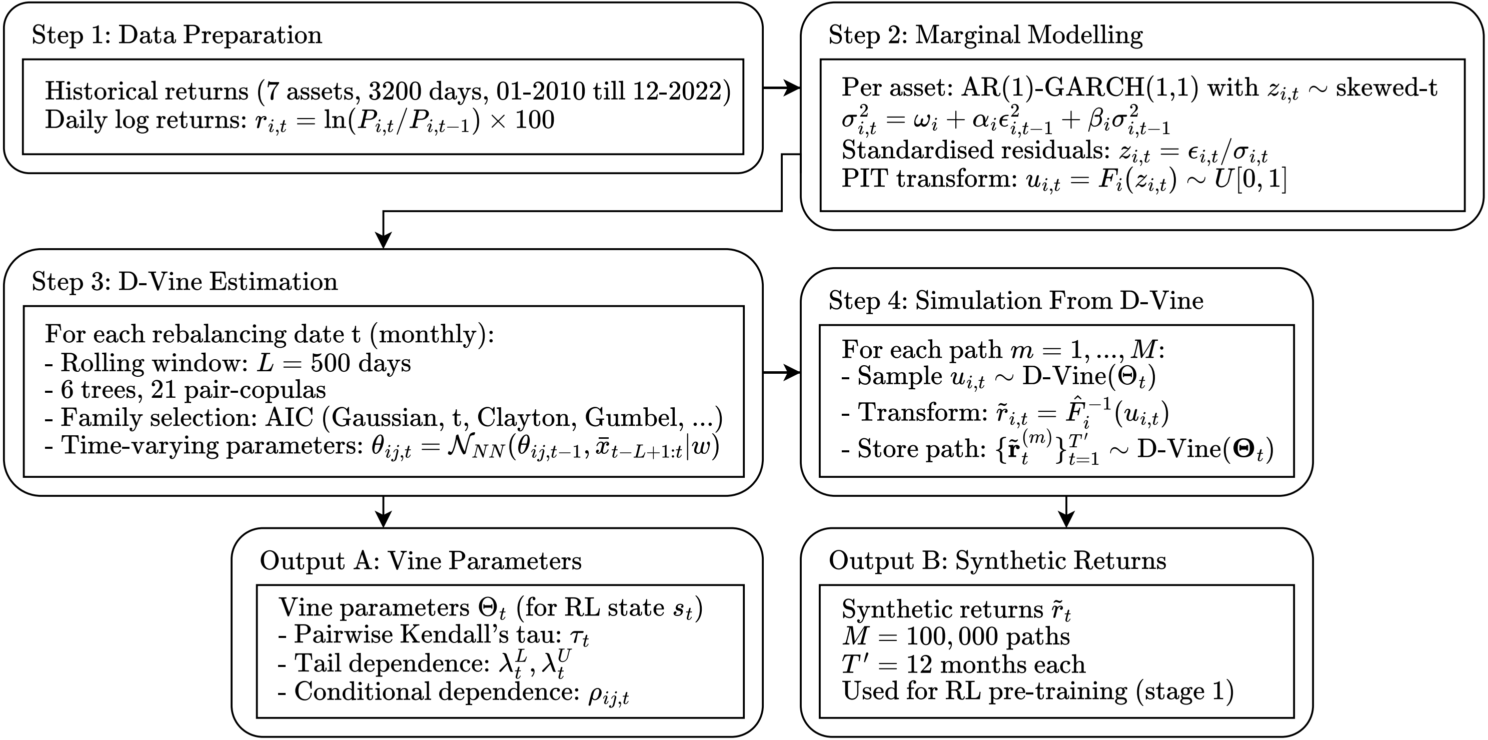
Step 1: Data Preparation
Historical returns (7 assets, 3200 days, 01-2010 till 12-2022)
Daily log returns: $r_{i,t} = \ln(P_{i,t}/P_{i,t-1}) \times 100$
Step 2: Marginal Modelling
Per asset: AR(1)-GARCH(1,1) with $z_{i,t} \sim$ skewed-t
$\sigma^2_{i,t} = \omega_i + \alpha_i \epsilon^2_{i,t-1} + \beta_i \sigma^2_{i,t-1}$
Standardised residuals: $z_{i,t} = \epsilon_{i,t}/\sigma_{i,t}$
PIT transform: $u_{i,t} = F_i(z_{i,t}) \sim U[0,1]$
Step 3: D-Vine Estimation
For each rebalancing date t (monthly):
- Rolling window: $L = 500$ days
- 6 trees, 21 pair-copulas
- Family selection: AIC (Gaussian, t, Clayton, Gumbel, ...)
- Time-varying parameters: $\theta_{ij,t} = \mathcal{N}_{NN}(\theta_{ij,t-1}, \bar{x}_{t-L+1:t} | w)$
Step 4: Simulation From D-Vine
For each path $m = 1, ..., M$:
- Sample $u_{i,t} \sim$ D-Vine(Θ_t)
- Transform: $\tilde{r}_{i,t} = \hat{F}_i^{-1}(u_{i,t})$
- Store path: $\{\tilde{\mathbf{r}}_t^{(m)}\}_{t=1}^{T'} \sim$ D-Vine($\boldsymbol{\Theta}_t$)
Output A: Vine Parameters
Vine parameters Θ_t (for RL state s_t)
- Pairwise Kendall's tau: τ_t
- Tail dependence: λ_t^L, λ_t^U
- Conditional dependence: $\rho_{ij,t}$
Output B: Synthetic Returns
Synthetic returns $\tilde{r}_t$
$M = 100,000$ paths
$T' = 12$ months each
Used for RL pre-training (stage 1)